

ABOUT ME

I'm **Kastab Garai**, a Cyber-Security researcher and Founder of the **Bug2Build Community**, a space that empowers hackers, builders, and students to explore technology.

My work in Responsible Disclosure has earned appreciation from:

IIT Bombay
IIT Roorkee
IIT Madras

Govt Of India
CGC University
TheSouledStore

Public Speaker
3x Hackathon
winner



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www.bug2build.in

Today's Agenda

- Open-Source Culture & Community Building
- Hackathons, Innovation & Problem Solving
- Technology Careers & Industry Exposure
- Student Leadership & skill development



HACKATHON, Innovation & Problem Solving

Origination of "Hackathon"

Hack – Cleverly
build

Marathon – Meaning a long,
sustained effort

The Birth

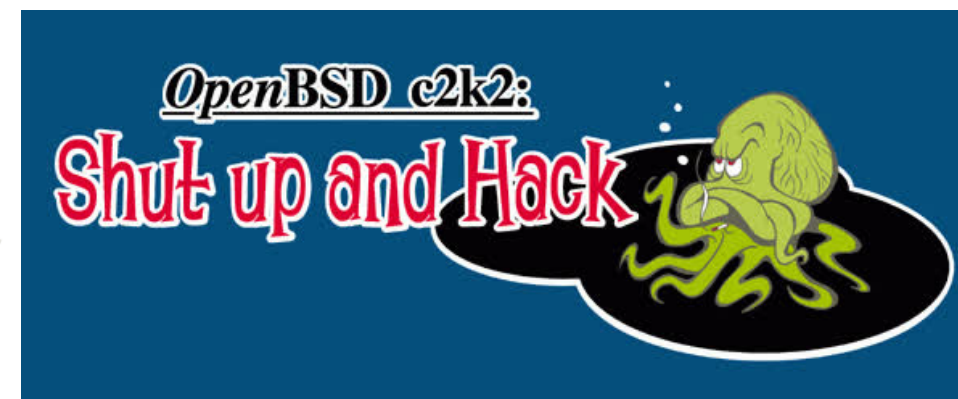
1. Hacker culture (1970s–1990s)

Programmers were already doing:

- *"hack days"*
- *all-night coding sessions*
- *collaborative problem solving*

2. Open-source community influence

As open-source projects grew (Linux, Apache, etc.), so did the idea of working together intensively to build software fast.



TYPES OF HACKATHON



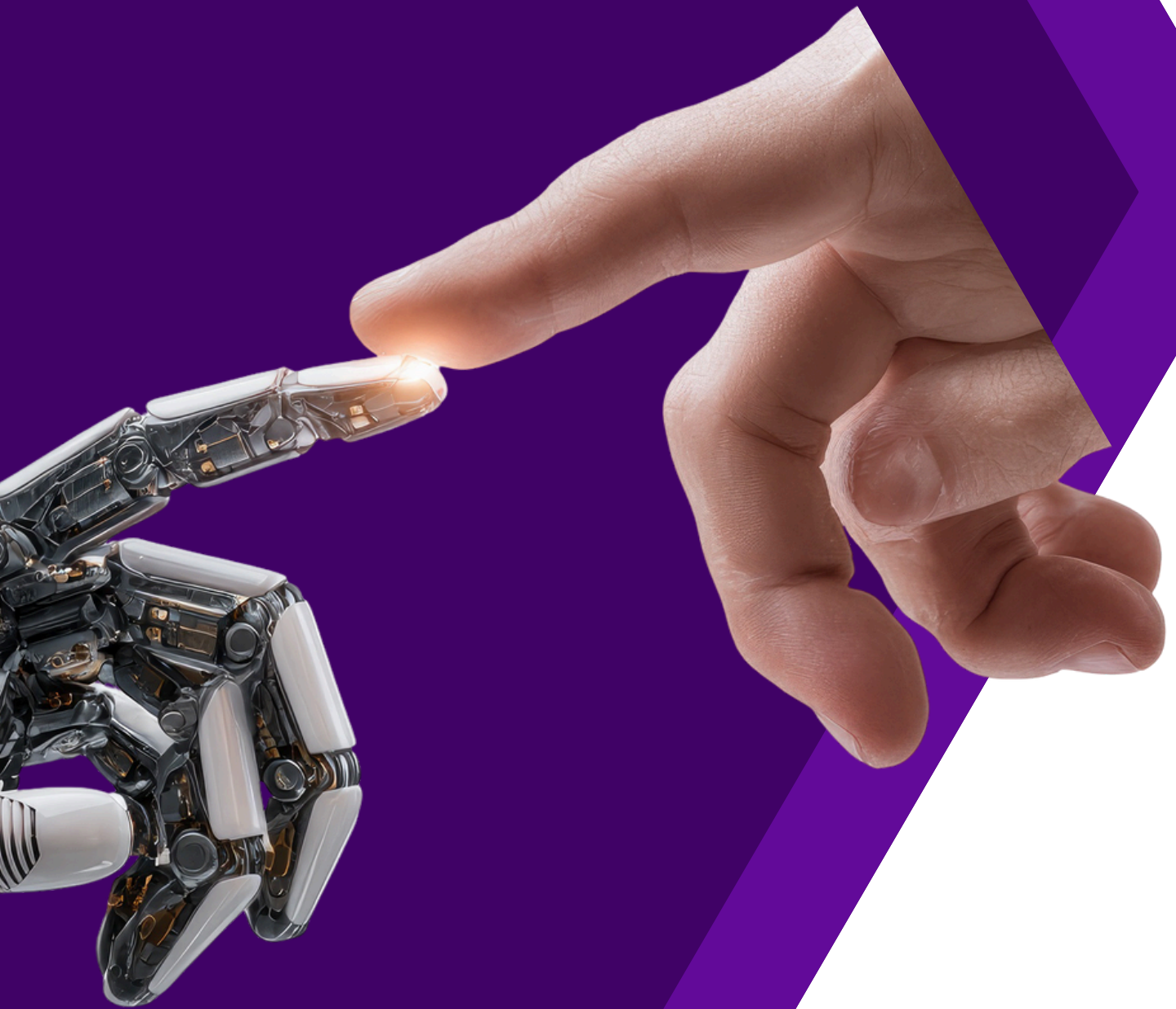
Student Hackathons

- Innovation Hackathons
- Government / Civic Hackathons
- IOT Hackathons
- Game Jams
- Ideation Hackathons
- CTF Hackathons



Corporate/Industry Hackathons

- Community / Open Hackathons
- Marathon Coding Competitions
- Virtual / Online
- Physical / In-Person
- Hybrid
- Startup Hackathons
- Mathematical Equations Hackathon



TYPES OF INNOVATION

- Technological Innovation
- Marketing Innovation
- Organizational Innovation
- Social Innovation
- Product Innovation
- Open Innovation
- Design Innovation
- Process Innovation
- Service Innovation
- Business Model Innovation

Problem Solving

Problem solving is the process of **identifying a problem** and finding an **effective solution** to overcome it.

Types of Problem Solving:

1. **Analytical** – uses logic and data to break down problems.
2. **Creative** – uses imagination to generate new ideas.
3. **Critical** – evaluates information and options before deciding.
4. **Algorithmic** – follows step-by-step methods or procedures.
5. **Collaborative** – solved through teamwork and discussion.



OPEN SOURCE CULTURE

India

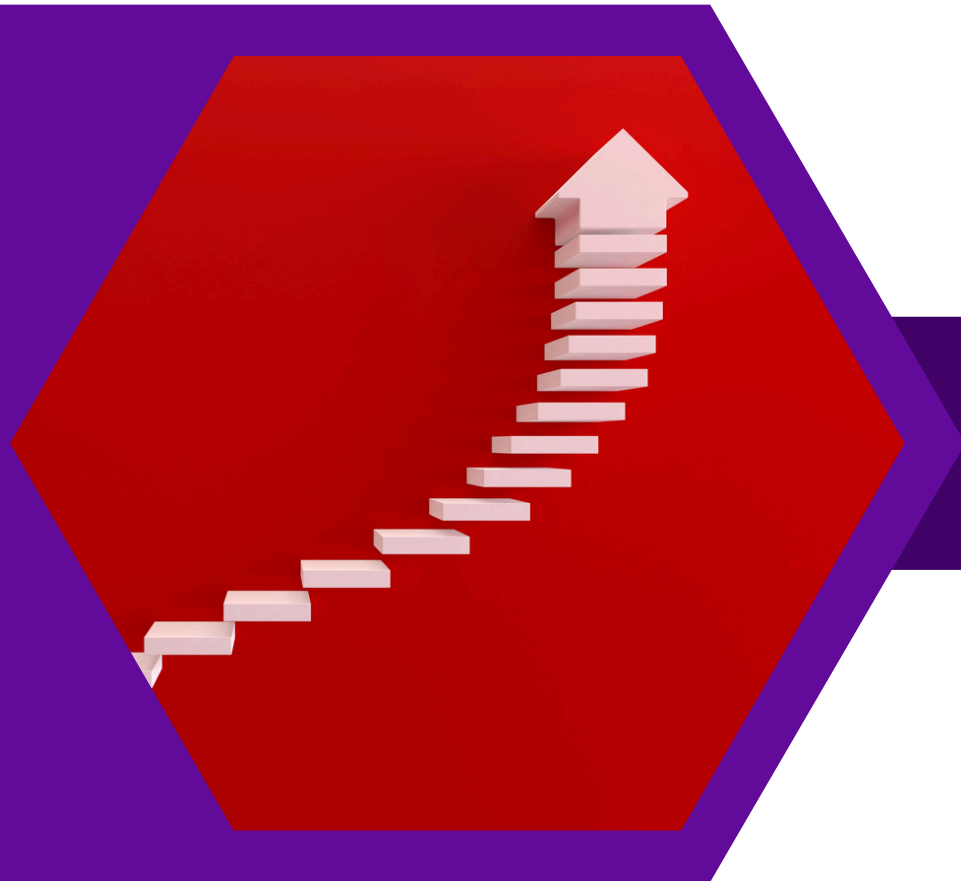
India Recently overtook the United States in the number of open source contributors on GitHub.
In 2025, India accounted for 14% of all new GitHub users and has the largest contributor base globally.
Projected to have 57+ million developers by 2030, making up about one in three new coders worldwide



United States

Traditionally the largest open source contributor in terms of code volume and influence.
Hosts many major foundations (e.g., Linux Foundation, Apache Foundation) and corporate contributors (Google, Microsoft).
Still leads in total contribution volume per developer even if India currently leads in headcount.





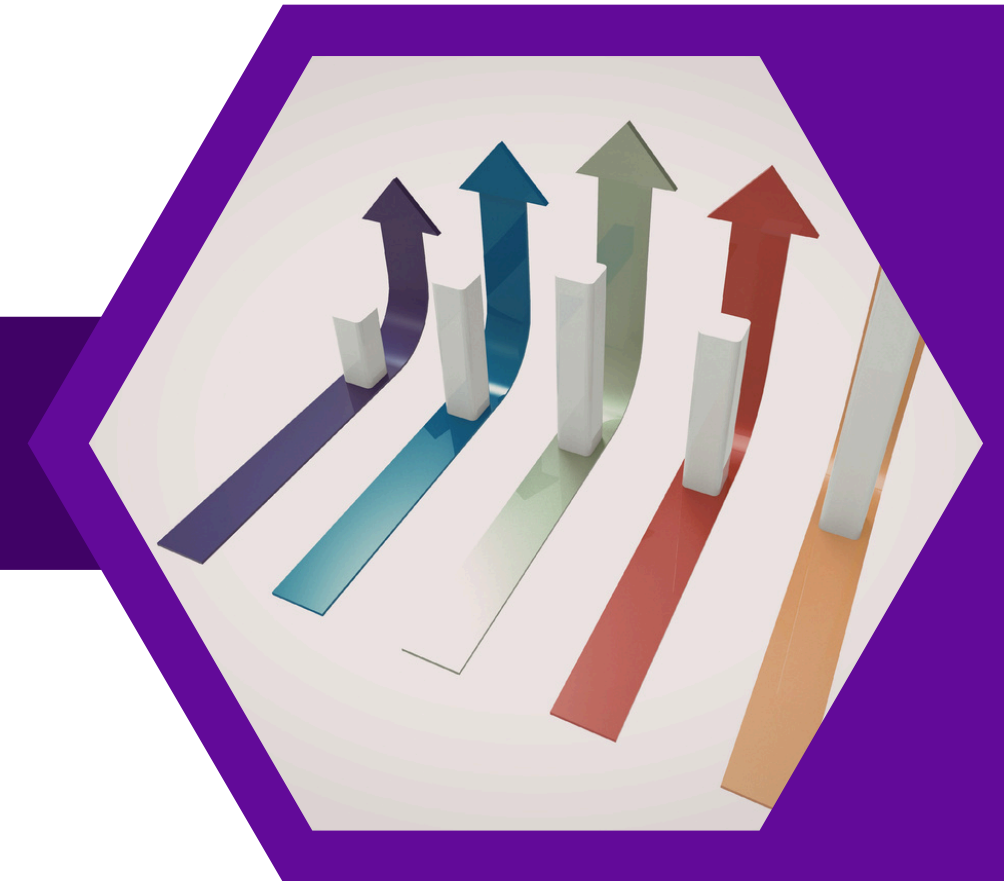
China

Rapid growth, open source developer pool grew over 10× since 2015. Together with India, China is a major engine of global open source development.



Brazil, Indonesia, Germany

Rising contributors with strong growth communities and increasing open source project creation. Reflect a global shift beyond traditional Western hubs.



Switzerland & Nordic Countries

High contributions per capita, especially in Europe, often stronger than larger nations when normalized by population.



STUDENT LEADERSHIP & SKILL DEVELOPMENT

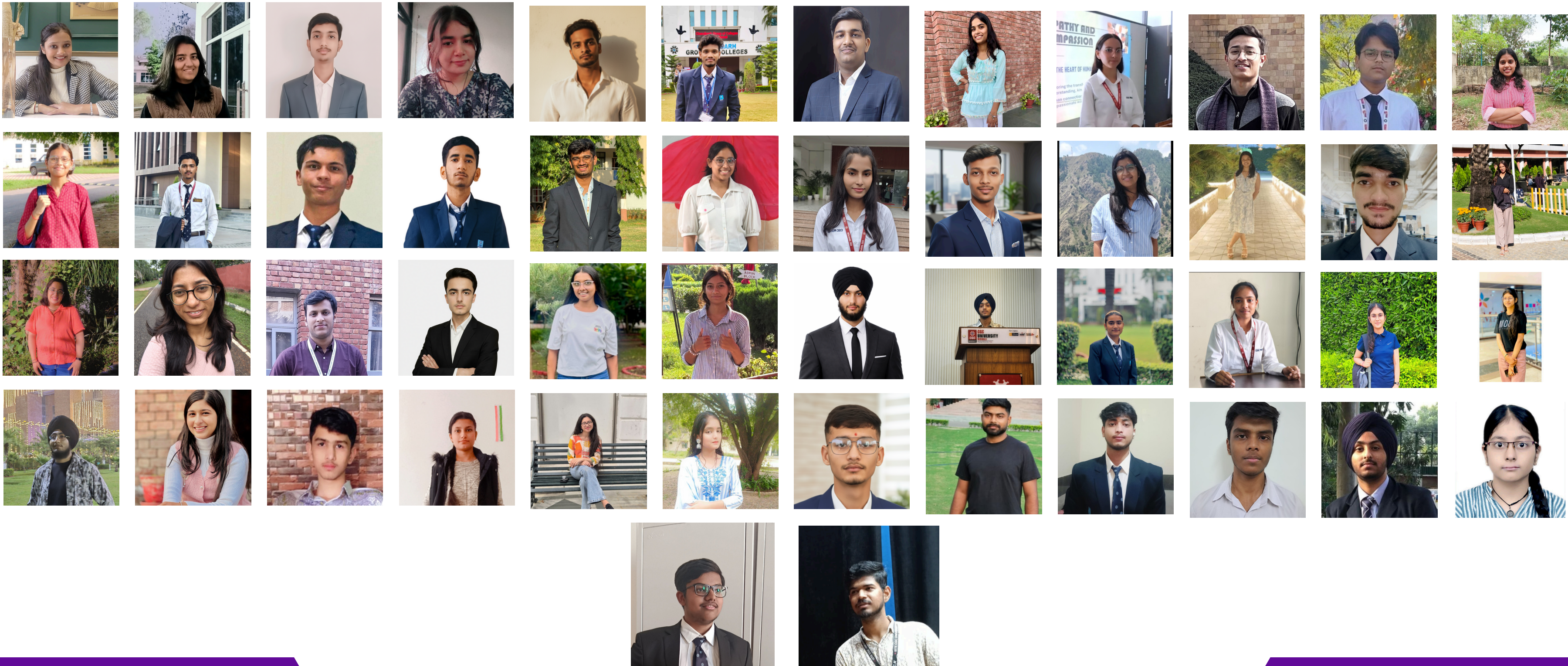
Student Leadership (What & Why)

- Student Leadership = ability to guide, influence, and take responsibility.
- Why it matters:
- Builds confidence
- Improves communication & teamwork
- Enhances problem-solving
- Prepares for real-world careers
- Encourages initiative & ownership

Skill Development (How & What to Build)

- Skills to develop:
- Communication
- Critical & creative thinking
- Time management
- Emotional intelligence
- Tech + digital skills
- Leadership & teamwork

MY TEAM MEMBERS



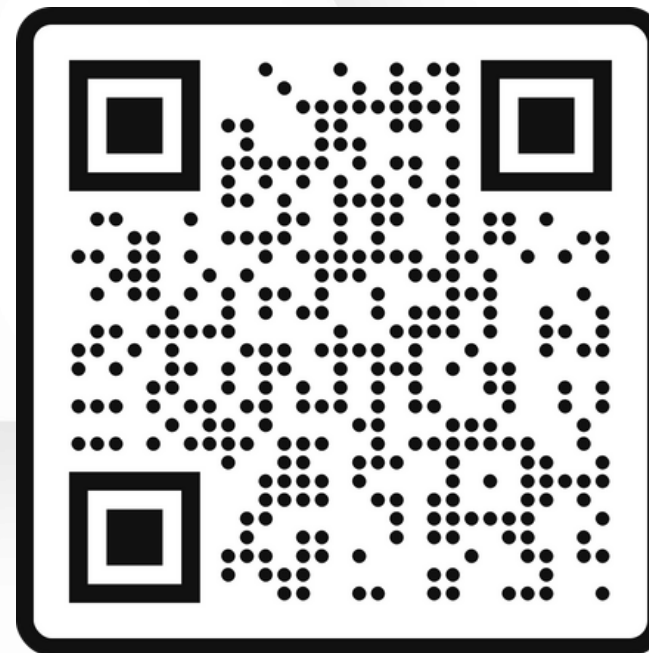
THANK YOU

TOGETHER, WE CREATE IMPACT

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